

A First-Principles Model of the Surface-EMG Forward Problem

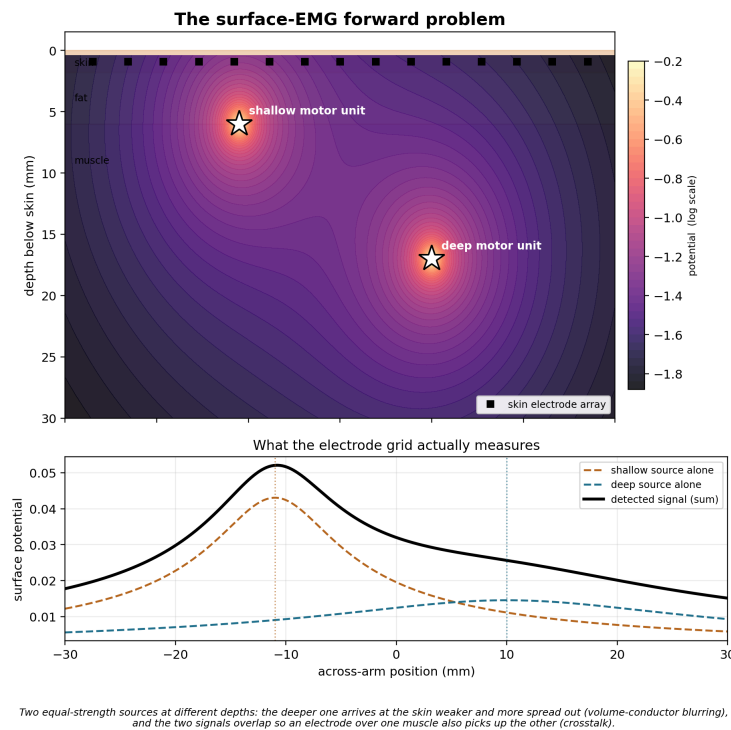
An imaging-derived 3D forearm model and a from-scratch simulation of how muscle electrical activity reaches skin electrodes. Santiago Benavides | June 2026

The problem

Surface electromyography (sEMG) reads muscle activity from skin electrodes, the signal behind modern prosthetic and brain-computer interface control. It is hard because the body acts as a volume conductor: a focal source deep in a muscle does not arrive cleanly at the skin. It is blurred by the intervening fat and skin, and it bleeds sideways onto electrodes meant for neighboring muscles (crosstalk). Understanding this forward path is the foundation for decoding the signal in reverse.

What I built

(1) An anatomically grounded 3D forearm model, segmented into four tissues (muscle, bone, fat, skin) from public-domain Visible Human cross-sectional imaging in 3D Slicer and rendered in Blender. (2) A from-scratch surface-EMG forward model in Python: a Rosenfalck action-potential current source propagated through a volume conductor to a grid of skin electrodes, reproducing the canonical detected waveform and quantifying crosstalk and depth-dependent blurring.



Two equal-strength sources at different depths: the deeper one arrives at the skin weaker and more spread out (volume-conductor blurring), and the two signals overlap so an electrode over one muscle also picks up the other (crosstalk).

Figure generated with Claude (Anthropic).

Figure 1. The forward problem (conceptual layered model). Two equal-strength motor units at different depths (top) produce very different surface signals (bottom): the deeper source arrives weaker and more spread out, and the two overlap, so one electrode reads both muscles (crosstalk).

Result

The model reproduces two key, real phenomena directly from the physics: depth/fat low-pass blurring (a deeper source of identical strength is markedly attenuated and smeared at the skin) and crosstalk between adjacent sources. These are precisely the effects that make HD-sEMG decoding difficult, here made visible and measurable from first principles.

Significance and next steps

This establishes a validated, fully understood pipeline from anatomy to simulated signal. Next: solve the field on the real segmented anatomy via finite elements (anisotropic muscle), and validate against measured sEMG, toward informing electrode-array design for a high-density sEMG gesture-decoding system.

Tools: 3D Slicer, Blender, Python (NumPy, Matplotlib). Basis: Rosenfalck (1969); Farina & Merletti (2001); Merletti & Farina (2016). Imaging: NLM Visible Human Project (public domain).